communication established the foundation for networked UAV operations (Kagawa, et al 2017).

- *GPS-based autonomous navigation*: The introduction of the global positioning system (GPS) allowed drones to follow predefined flight paths and return automatically to the launch point. GPS integration enabled waypoint navigation, hover stability, and precise position tracking, which reduced pilot workload and improved mission repeatability.
- *Electronic flight stabilization*: Equipped with inertial measurement units (IMUs) and gyroscopic sensors, these drones could automatically correct roll, pitch, and yaw deviations. This stabilization system ensured smooth flight even in turbulent conditions and was one of the first steps toward autopilot technology.
- *Digital telemetry and real-time feedback*: The digital revolution introduced telemetry systems capable of transmitting video, sensor data, and system diagnostics to the ground station. This advancement allowed for real-time situational awareness, critical for both reconnaissance and scientific missions.
- *Payload integration and mission flexibility*: Second-generation drones were designed to accommodate a range of modular payloads, including cameras, infrared sensors, and scientific instruments. This flexibility made UAVs suitable for diverse missions across military, environmental, and commercial domains.

3.3.2 Major components of second-generation drones

The second generation of drones introduced a significant leap in unmanned aerial vehicle (UAV) engineering through the integration of advanced electronic, aerodynamic, and navigational systems. Unlike the simple, manually controlled drones of the first generation, these UAVs were